

*A COMPREHENSIVE
LITERATURE REVIEW AND
FUTURE RESEARCH
OPPORTUNITIES IN
OPTIMIZATION APPROACHES
IN PROJECT PORTFOLIO
MANAGEMENT UNDER
UNCERTAINTY AND IN
DYNAMIC ENVIRONMENTS*

Sarah Wolberg

*Stevens Institute of Technology | Castle Point on
Hudson, Hoboken, NJ 07030*

AGENDA

- Introduction
- Research Objectives
- Key Studies Reviewed
- Analysis For Each Study (e.g., Synopsis, Model Components, Model Objective, Gaps in Study, and Opportunities for Future Research)
- Similarities Across Studies
- Differences Across Studies
- Future Research Opportunities and Possible Areas of Focus
- Overall Recommendations
- Conclusions
- Acknowledgments
- Q&A
- References

INTRODUCTION

Background on Project Portfolio Management (PPM): PPM faces increasing challenges in dynamic environments with fluctuating demands, evolving project scopes, and uncertainties in resource availability.

Problem Statement: Traditional models often fail to handle dynamic environments effectively.

Key Insights: Effective PPM should integrate dynamic decision-making based on project risks, enabling continuous resource adaptation as new information becomes available.

Purpose of the Review: This paper synthesizes methodologies, compares models, identifies research gaps, and proposes directions for future work.

RESEARCH OBJECTIVES

Objective 1: Review and compare existing research on PPM methodologies under uncertainty.

Objective 2: Analyze the strengths and limitations of optimization models (e.g., multistage programming, risk-averse methods).

Objective 3: Identify similarities and differences across studies.

Objective 4: Propose future research opportunities to improve model adaptability, multi-objective optimization, and computational efficiency.

Objective 5: Outline how these insights will influence further research.

KEY STUDIES REVIEWED

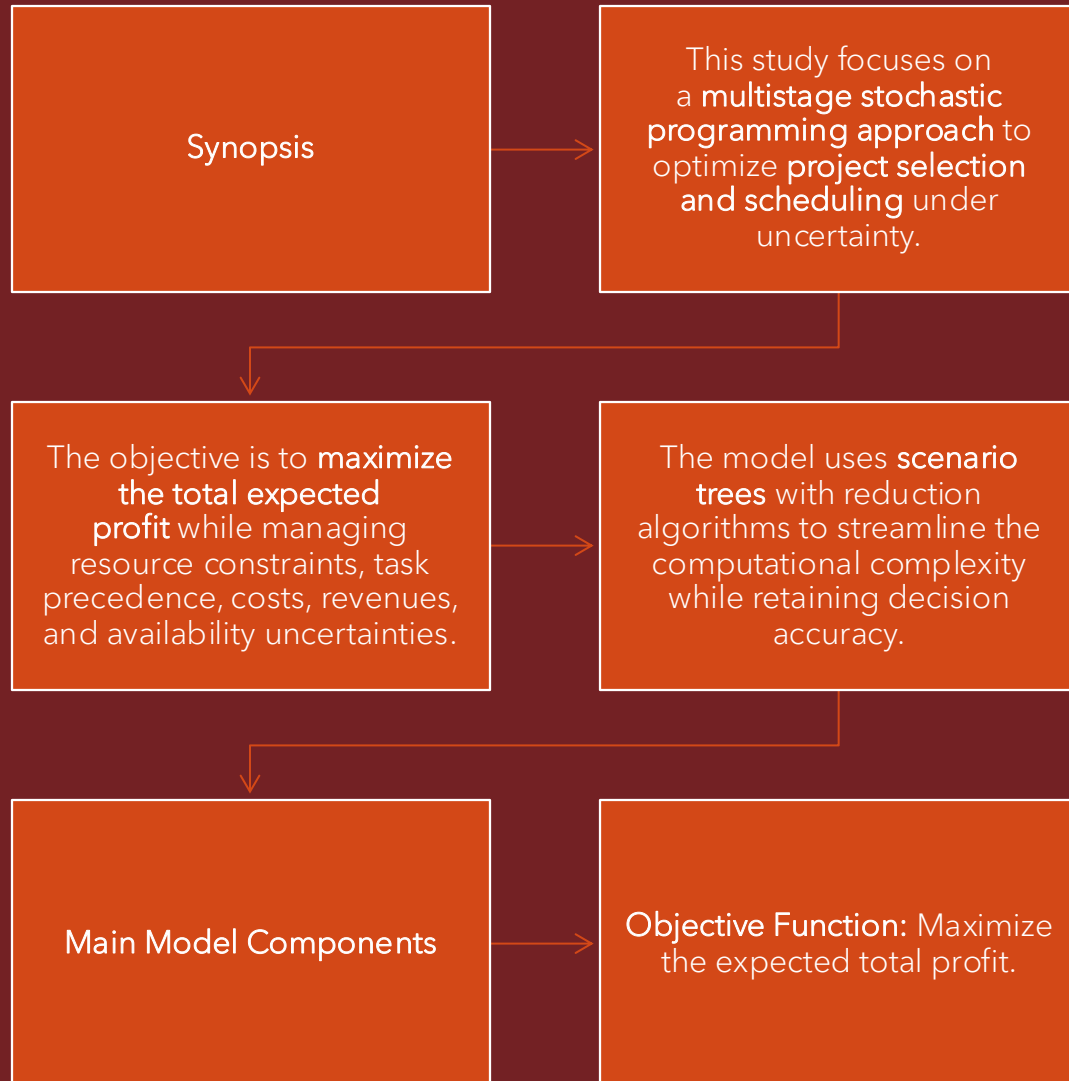
1. "A Multistage Stochastic Programming Approach in Project Selection and Scheduling" (International Journal of Advanced Manufacturing Technology).
2. "Stochastic Project Management: Multiple Projects with Multi-skilled Human Resources" (Journal of Scheduling).
3. "Robust and Resilience Budget Allocation for Projects with a Risk-averse Approach: A Case Study in Healthcare Projects" (Computers & Industrial Engineering).
4. "Risk-averse Supply Chain for Modular Construction Projects" (Automation in Construction).
5. "A clustering-based review on project portfolio optimization methods" (*International Transactions in Operational Research*).
6. "Integrated Framework for Project Portfolio Planning Management Under Resource Constraints" (Master's thesis, Stevens Institute of Technology).
7. "Robust optimization for hurricane preparedness" (*International Journal of Production Economics*).
8. "Optimization of renewable energy project portfolio selection using hybrid AIS-AFS algorithm in an international case study" (*Scientific Reports*).

KEY STUDIES REVIEWED (CONTINUED)

Study	Source	Focus
1. Multistage Stochastic Programming	<i>International Journal of Advanced Manufacturing Technology</i>	Project selection and scheduling.
2. Stochastic Project Management	<i>Journal of Scheduling</i>	Scheduling projects with multi-skilled resources.
3. Robust Budget Allocation	<i>Computers & Industrial Engineering</i>	Risk-averse budget allocation in healthcare.
4. Risk-Averse Supply Chain	<i>Automation in Construction</i>	Modular construction logistics under uncertainty.
5. Clustering-Based Optimization	<i>International Transactions in Operational Research</i>	Optimization methods review.
6. Integrated Framework	<i>Master's Thesis, Stevens Institute of Technology</i>	Portfolio management under resource constraints.
7. Robust Hurricane Preparedness	<i>International Journal of Production Economics</i>	Disaster response optimization.
8. Renewable Energy Selection	<i>Scientific Reports</i>	Project portfolio optimization using hybrid models.

**ANALYSIS (E.G., SYNOPSIS, MODEL COMPONENTS,
MODEL OBJECTIVE, GAPS IN STUDY, AND
OPPORTUNITIES FOR FUTURE RESEARCH)**

“MULTISTAGE STOCHASTIC PROGRAMMING APPROACH IN PROJECT SELECTION AND SCHEDULING”



Main Model Components

Objective Function: Maximize the expected total profit.

Objective Function:

$$\text{Maximize } \sum_{n \in N} \pi_n \sum_{i=1}^I \sum_{j=1}^{J_i} P_{ij}^n X_{ij}^n$$

Objective: The objective is to maximize the total expected profit across all projects, tasks, and scenarios. Each term in the equation contributes to the overall expected value by accounting for different scenarios.

The summation over $n \in N$, n represents a node in a scenario tree, which is part of a set of scenarios N , and each node n represents a possible outcome at a given stage in the scenario tree.

π_n is the probability associated with scenario n . It represents the likelihood of that scenario occurring.

P_{ij}^n This represents the profit (or present value) of task j from project i in scenario n . Since the profit depends on the scenario, it is indexed by n .

X_{ij}^n is a binary decision variable (0 or 1) indicating whether task j in project i is selected in scenario n . $X_{ij}^n = 1$, task j is started; if $X_{ij}^n = 0$, it is not started.

“MULTISTAGE STOCHASTIC PROGRAMMING APPROACH IN PROJECT SELECTION AND SCHEDULING”

Key Constraints

Task Precedence Constraint

$$d_{ij}X_{i(j+1)}^n \leq \sum_{\gamma=1}^{t(n)-1} (t(n) - l)X_{ij}^{n-\gamma}; l < (n), \forall n \in N$$

Resource Constraint

$$\sum_{i=0}^I \sum_{j=0}^{J_i} r_{ijk}^n \sum_{v=0}^{d_{ij}-1} X_{ij}^{n-v} \leq R_k^n \forall k = 1, \dots, K, n \in N$$

Task Selection Constraint

$$\sum_{l=1}^{t(n)} X_{ij}^{n-\gamma} \leq 1; \forall i, \forall j = 1, \dots, J_i, \forall n \in N$$

Case Study:

Application: R&D company scheduling projects over 12 months.

Outcome: Achieved optimal profit of **107 units** using scenario tree reduction (MATLAB implementation) and reduces scenarios (e.g., 2,400 nodes to 84) to lower computational complexity while maintaining accuracy.

Key Insight: The model effectively manages uncertainty but requires improvements for real-world adaptability.

“MULTISTAGE STOCHASTIC PROGRAMMING APPROACH IN PROJECT SELECTION AND SCHEDULING”

- While the study demonstrates the power of stochastic programming for project scheduling under uncertainty, there are numerous opportunities for improving the model by incorporating more complex, realistic constraints and objectives.
- Future research could focus on making the model more adaptive, flexible, and capable of handling multiple objectives and dynamic changes.

Gaps and Opportunities for Future Research

Identified Gaps In Study	Description	Opportunities for Future Research
Expansion into Non-Financial Parameters	The study focuses primarily on resource availability in terms of financial and human resources. Future research could consider other non-financial uncertainties, such as technology changes, market demands, or project interdependencies.	Incorporating non-financial parameters like risk assessment, technology advancements and competitor actions would provide a more comprehensive decision-making framework.
Incorporation of Time-Dependent Resource Constraint	The current model assumes that resources (budget, staff) are available at specific points in time and do not change dynamically during the project. However, real-world projects often face time-dependent variations in resource availability, such as sudden budget cuts or shifts in personnel.	Future models could introduce dynamic resource constraints, where resource availability fluctuates within a project based on external factors (e.g., company funding cycles or staff attrition).
Multi-Objective Optimization	The study focuses on a single objective: maximizing profit. However, many real-world scenarios require a multi-objective approach, balancing not just profit but also factors like project completion time, risk minimization, and social impact.	Future research could explore multi-objective stochastic programming, where objectives like profit, project completion time, and risk mitigation are considered simultaneously. This could provide a more robust decision-making tool for complex projects with competing goals.
Project Cancellations and Flexibility	The current model does not account for the possibility of cancelling or delaying projects due to unfavorable scenarios. In practice, organizations often need the flexibility to cancel projects or pause tasks when resources are severely constrained.	Introducing flexibility in project selection, allowing for the cancellation or rescheduling of projects in response to scenario outcomes, would provide more realistic solutions.
Real-Time Decision Making	The model operates in a batch mode, where all decisions are made at the beginning of the planning period. However, in many real-world cases, decisions must be updated in real-time as new information becomes available (e.g., new budget updates or unforeseen delays).	Developing real-time stochastic decision-making models that allow for continuous updates and adaptive scheduling as new data arrives would improve the practicality of the model.

"STOCHASTIC PROJECT MANAGEMENT: MULTIPLE PROJECTS WITH MULTI-SKILLED HUMAN RESOURCES"

Synopsis - The study titled "Stochastic Project Management: **Multiple Projects with Multi-Skilled Human Resources**" presents two stochastic optimization approaches to solve project scheduling and personnel staffing problems under **uncertain work time demands**. It extends the deterministic model by Heimerl and Kolisch by incorporating **stochasticity in work package processing times**.

The focus is on scheduling **multiple projects** that share human resources with **heterogeneous skill levels**, where the main challenge is balancing **internal resource capacity** and costly **external resources** when internal demand exceeds availability.

The two proposed solution approaches are:

1. **Matheuristic Approach:** Combines a metaheuristic for scheduling projects (using **Iterated Local Search (ILS)** with variable neighborhood descent) and the **Frank-Wolfe algorithm** to optimize staffing.
2. **Sample Average Approximation (SAA):** Approximates the stochastic problem by solving multiple deterministic instances using **mixed-integer programming** with sampled scenarios.

The model was tested on synthetically generated data inspired by an IT department of a large semiconductor company. Results showed:

- The **Matheuristic approach** performed robustly for **small to medium-sized instances**.
- The **SAA approach** worked under specific conditions but struggled with larger problem instances due to scalability issues.

"STOCHASTIC PROJECT MANAGEMENT: MULTIPLE PROJECTS WITH MULTI-SKILLED HUMAN RESOURCES"

Main Model Components

Objective: Minimize the expected cost of **external resources** when demand exceeds internal capacity. Note: External costs are used when the demand for a particular skill in a project exceeds the capacity of internally scheduled capacity or resources [2].

Objective Function

$$\text{Minimize } \sum_{p \in P} \sum_{t=ES_p}^{LF_p} \sum_{s \in S} c_s^e \mathbb{E} \left(\left[\sum_{q \in \tau_{pt}} D_{psq} z_{pqt} - \sum_{k \in K_s} x_{ptsk} \right]^+ \right)$$

Constraints

Activity Scheduling Constraint

$$\sum_{t=ES_{pq}}^{LS_{pq}} z_{pqt} = 1 \quad \forall p \in P, q = 1, \dots, d_p$$

Precedence Constraint

$$\sum_{t=ES_{pq}}^{LS_{pq}} t z_{pqt} < \sum_{t=ES_{p,q+1}}^{LS_{p,q+1}} t z_{p,q+1,t} \quad \forall p \in P, q = 1, \dots, d_p - 1$$

Resource Capacity Constraint

$$\sum_{p \in P} \sum_{s \in S} \frac{1}{n_{sk}} x_{ptsk} \leq a_{kt} \quad \forall k \in K, t = 1, \dots, T, p \in P$$

Both approaches share the same foundational equations (objective function, scheduling constraints, and resource capacity constraints), but they differ in how they solve the problem. The matheuristic approach uses an iterative local search and convex optimization, while SAA relies on scenario-based sampling and deterministic optimization.

"STOCHASTIC PROJECT MANAGEMENT: MULTIPLE PROJECTS WITH MULTI-SKILLED HUMAN RESOURCES"

Equations in Matheuristic Approach

Staffing Subproblem (Convex Optimization using Frank-Wolfe Algorithm)

$$\text{Minimize } \sum_{p \in P} \sum_{t=ES_p}^{LF_p} \sum_{s \in S} c_s^e \mathbb{E} \left(\left[\sum_{q \in \tau_{pt}} D'_{psq} - \sum_{k \in K_s} x_{ptsk} \right]^+ \right)$$

1. The matheuristic approach solves the problem by breaking it down into two subproblems: project scheduling and staffing.
2. The project scheduling part is handled by the iterated local search (ILS) metaheuristic, while the staffing problem is solved using the Frank-Wolfe algorithm for convex optimization [2].

Sample Average Approach

Deterministic Equivalent problem

$$\text{Minimize } \frac{1}{N} \sum_{n=1}^N \sum_{p \in P} \sum_{t=ES_p}^{LF_p} \sum_{s \in S} c_s^e y_{pts}^{(n)}$$

1. In the Sample Average Approximation (SAA) approach, the stochastic problem is approximated by generating several sample scenarios of the random variables (i.e., different realizations of work package demands). The expected cost is then approximated by averaging over these scenarios.
2. The objective is to minimize the average cost over all scenarios. The deterministic equivalent problem is a mixed-integer linear program (MILP) that can be solved using solvers.

" STOCHASTIC PROJECT
MANAGEMENT: MULTIPLE
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SKILLED HUMAN RESOURCES "

Results

Overall, the Matheruist model experimental results performed better for small and medium-sized test instances. The SAA model failed to return any results. The results also proved that some managerial insights are valid regarding uncertainty on required efforts.

Gaps and Opportunities for Future Research

Identified Gaps In Study	Description	Opportunities for Future Research
Limited Consideration of Resource Substitutability	The study focuses primarily on the substitutability of internal resources with external resources in cases of insufficient internal capacity. However, it does not explore within-skill or between-skill substitutions of internal resources by other internal resources, which is a common recourse action during work overload. This limitation suggests a gap in addressing how organizations can optimize internal resource flexibility to reduce external costs further.	Future research should develop models that incorporate these forms of reactive planning and explore solution techniques to handle the complexity of this extended framework.
Risk-Neutral Assumptions	The current model operates under the assumption of risk neutrality, which is appropriate for multi-project management with relatively small or medium-sized projects. However, in cases where larger projects or higher-risk scenarios are involved, risk-neutral optimization may not be suitable.	Future studies should investigate risk-averse optimization approaches, which account for specific risks associated with large, complex projects. Incorporating risk aversion would offer a more conservative and practical planning strategy in high-stakes environments.

" ROBUST AND RESILIENCE BUDGET ALLOCATION FOR PROJECTS WITH A RISK-AVERSE APPROACH: A CASE STUDY IN HEALTHCARE PROJECTS"

Synopsis

This study introduces the **3RBAP (Robust, Resilience, and Risk-Averse Budget Allocation)** model to optimize **budget allocation** for projects under uncertainty.

Focus: Large-scale **healthcare projects** where timely execution is critical.

Objective: Maximize project progress by balancing budget allocation to mitigate uncertainty and worst-case scenarios using robust optimization.

Objective Function: Maximize weighted expected and minimum progress.

$$\text{Maximize } Z = (1 - \beta)\Gamma + \beta\Gamma$$

Subject to:

$$\Gamma = \sum_s p p_s \Gamma_s$$

Constraints

Progress constraints

$$\Gamma = \min_s (\Gamma'_s), \quad \forall i, s$$

$$\Gamma'_s = \frac{\sum_i f f_{is}}{|I|}, \quad \forall s$$

$$f f_{is} = w_i k k_{is}, \quad \forall i, s$$

" ROBUST AND RESILIENCE BUDGET ALLOCATION FOR PROJECTS WITH A RISK-AVERSE APPROACH: A CASE STUDY IN HEALTHCARE PROJECTS"

Constraints (Continued)

Budget Constraints

$$kk_{is} = \frac{x_{is}}{bac_{is}}, \text{Type 1 Linear}, \forall i, s$$

Or

$$kk_{is} = \left(\frac{x_{is}}{bac_{is}}\right)^n (1 - p_{is}), n \geq 0, \quad \text{Type 2 Nonlinear Degree}, \forall i, s$$

Or

$$kk_{is} = \left(\frac{e^{-x_{is}} - 1}{e^{-bac_{is}} - 1}\right)^n (1 - p_{is}), n \geq 0, \quad \text{Type 3 Nonlinear Exponential}, \forall i, s$$

Or

$$kk_{is} = (\log_a((a - 1)\left(\frac{x_{is}}{bac_{is}} + 1\right)))^n (1 - p_{is}), n \geq 0, a > 0, \quad \text{Type 4 Nonlinear Logarithm}, \forall i, s$$

Or

$$kk_{is} = \left(\sin\left(\frac{\pi}{2} \frac{x_{is}}{bac_{is}}\right)\right)^n (1 - p_{is}), n \geq 0, \quad \text{Type 5 Nonlinear Sine}, \forall i, s$$

$$x_{is} \leq bac_{is}, \forall i, s$$

" ROBUST AND RESILIENCE BUDGET ALLOCATION FOR PROJECTS WITH A RISK-AVERSE APPROACH: A CASE STUDY IN HEALTHCARE PROJECTS "

Results

Performance of 3RBAP:

- 1. **17.07% reduction** in the progress function compared to standard methods.
- 2. Increasing the **conservative coefficient (β) to 50%** led to a **7.32% reduction** in progress function with minimal computational overhead.
- 3. Switching to **exponential and sine functions** improved project outcomes by **41.46%**.

Sensitivity Analysis:

- 1. Lowering the **resiliency coefficient (δ)** disrupted budget allocation, leading to poorer progress outcomes.
- 2. **Scalability:** As problem size increases, computation time grows, but progress function performance improves.

Gaps and Opportunities for Future Research

Identified Gaps	Description	Opportunities for Research
Expanding Risk Models	Focused on budget uncertainty alone.	Incorporate workforce availability, regulations, and market volatility .
Dynamic Resource Reallocation	Static budget allocation assumptions.	Explore real-time resource allocation models to adapt to dynamic conditions.
Risk-Neutral Optimization	Assumes risk neutrality.	Develop risk-averse decision frameworks for larger, high-risk projects.
Computational Complexity	SAA increases time for large-scale problems.	Use metaheuristic or approximation algorithms to improve scalability.

"RISK-AVERSE SUPPLY CHAIN FOR MODULAR CONSTRUCTION PROJECTS"

Synopsis

- This study addresses **supply chain uncertainties** in **modular construction projects** using a **Robust Optimization (RO)** approach.

Challenge: Logistics complexities caused by factors such as weather delays, labor disruptions, and equipment malfunctions.

Objective: Minimize **operational costs** (manufacturing, transportation, inventory, and warehouse setup) while accounting for **risk aversion**.

Key Features:

- Optimize **warehouse locations**, **production schedules**, **transportation routes**, and **inventory management** under uncertain demand.
- Compare the RO approach with **two-stage stochastic programming (2SP)**.



"RISK-AVERSE SUPPLY CHAIN FOR MODULAR CONSTRUCTION PROJECTS"

Model Components

Objective Function: Minimize total costs under uncertainty.

$$\text{Minimize } ETC + \lambda \sum_{s \in S} P_s (TC_s - ETC + 2\theta_s) + \omega \sum_{s \in S} P_s \delta_s$$

Constraints:

Fixed Variable Production Costs (Fixed Production Costs (FPC) and Variable Production Costs (VPC)

$$FPC = MF \left(\sum_{a \in A} y_a + \sum_{p \in P} \hat{y}_p \right)$$

$$VPC = \sum_{a \in A} \sum_{j \in J} m_{aj} \cdot MV_j + \max \left(\left[m_{aj} - \frac{MMR_j}{2} \right], 0 \right) \cdot MV_j + \sum_{p \in P} \sum_{j \in J} \hat{m}_{pj} \cdot MV_j + \max \left(\hat{m}_{pj} - \frac{MMR_j}{2}, 0 \right) \cdot MV_j$$

$$\widehat{IC} = \sum_{p \in P} \sum_{z \in Z} \sum_{j \in J} i_{pzj}^W \cdot V_j \cdot SC^W$$

Total Inventory Costs

$$IC_s = \sum_{a \in A} \sum_{j \in J} i_{aj}^W \cdot V_j \cdot SC^F + \sum_{a \in A} \sum_{z \in Z} \sum_{j \in J} i_{pzj}^W \cdot V_j \cdot SC^W + \max(i_{saj}^W) \cdot V_j \cdot SC^C$$

Transportation Costs After Construction Work Starts

$$\widehat{TRC}^{\overrightarrow{FW}} = \sum_{p \in P} \sum_{z \in Z} \sum_{j \in J} (\widehat{t}_{pzj}^{\overrightarrow{FW}} / L_j^{\overrightarrow{FW}}) \cdot D_z^{\overrightarrow{FW}} \cdot TC_{pj}^{\overrightarrow{FW}}$$

Transportation Costs Between the Factory and Warehouse for Initial Inventory Preparedness Period

$$TRC^{\overrightarrow{FW}} = \sum_{a \in A} \sum_{z \in Z} \sum_{j \in J} (\widehat{t}_{azj}^{\overrightarrow{FW}} / L_j^{\overrightarrow{FW}}) \cdot D_z^{\overrightarrow{FW}} \cdot TC_{aj}^{\overrightarrow{FW}}$$

Transportation Costs Between the Warehouse and Construction Sites at the Beginning of Construction

$$TRC^{\overrightarrow{WC}} = \sum_{a \in A} \sum_{z \in Z} \sum_{j \in J} (\widehat{t}_{azj}^{\overrightarrow{FW}} / L_j^{\overrightarrow{FW}}) \cdot (D_z^{\overrightarrow{FW}} \cdot TC_{aj}^{\overrightarrow{FW}} + LP_z)$$

Fixed Costs on the Construction Site

$$SFC_s = SF \cdot DD_s$$

"RISK-AVERSE SUPPLY CHAIN FOR MODULAR CONSTRUCTION PROJECTS"

Constraints:

Extension Costs for Each Scenario

$$EX_s = \max (x - DD_s, 0) \cdot SF$$

Warehouse Costs by Location

$$WO = \sum_{z \in Z} e_z WEC_z$$

Measurement of Model Infeasibility

$$\delta_s = \sum_{a \in A} \sum_{j \in J} \min (i_{saj}^C, 0) \cdot + \sum_{a \in A} \max \left[\sum_{z \in Z} \max (i_{saj}^C, 0) - N, 0 \right]$$

a. Sum of Total Cost for each Scenario

$$TC_s = FPC + VPC + \widehat{IC} + IC_s + \widehat{TRC}^{\overrightarrow{FW}} + TRC^{\overrightarrow{FW}} + TRC^{\overrightarrow{WC}} + SFC_s + EX_s + WO$$

b. Expected Value of the Total Cost(s)

$$ETC = FPC + VPC + \widehat{IC} + \widehat{TRC}^{\overrightarrow{FW}} + TRC^{\overrightarrow{FW}} + TRC^{\overrightarrow{WC}} + WO + \sum_{s \in S} P_s (IC_s + SFC_s + EX_s)$$



"RISK-AVERSE SUPPLY CHAIN FOR MODULAR CONSTRUCTION PROJECTS"

Key Results

Effectiveness of RO Model:

- **58.53% reduction** in cost variance compared to the two-stage stochastic programming (2SP) model.
- **25.91% reduction** in the standard deviation of operational costs.
- RO minimized **maximum cost overruns** by £6,380 in worst-case scenarios.

Metric	RO Model	2SP Model
Cost Variance Reduction	58.53%	N/A
Standard Deviation Reduction	25.91%	N/A
Expected Cost Savings	Slightly higher	0.38% better
Risk Management (Worst-Case)	Significant	Mod

Warehouse and Production Optimization:

- Selected **optimal warehouse location (Z1)** to minimize delays and improve responsiveness.
- Optimal production duration:
 - **6 weeks** for initial inventory preparation.
 - **3 weeks** for production during the assembly phase.

"RISK-AVERSE SUPPLY CHAIN FOR MODULAR CONSTRUCTION PROJECTS"

Gaps in Study and Opportunities for Future Research

Identified Gaps	Description	Opportunities for Research
Upstream Supply Chain Risks	Focused on downstream logistics uncertainties.	Include manufacturing delays and raw material shortages for a more holistic supply chain model.
Narrow Industry Focus	Limited to modular construction projects.	Apply the model to other sectors like healthcare, energy, or manufacturing .
Advanced Risk Aversion	General risk-aversion approach used.	Develop sophisticated risk metrics tailored to varying risk tolerance levels.
Computational Complexity	Model scalability for large projects.	Explore metaheuristic algorithms to improve computational efficiency.



“A CLUSTERING-BASED REVIEW ON PROJECT PORTFOLIO OPTIMIZATION METHODS”

Synopsis

- This paper offers a **comprehensive review** of **project portfolio optimization (PPO) methods** using a **clustering-based approach** to group and analyze research trends.
- **Scope:**
 - Reviewed **3,183 publications**, narrowed to **298 key papers** grouped into **24 clusters**.
 - Identified dominant methodologies and their applications in Project Portfolio Management (PPM).

Focus Areas

- **Metaheuristics, fuzzy modeling, multi-criteria decision analysis (MCDA), and hybrid models.**

Key Insight

Optimization methods have evolved to address complexities like **resource constraints, uncertainty, interdependencies, and multi-objective decision-making.**

Main Categories of PPO Methods

- 1. Metaheuristics:** Scalable and adaptable for large, constrained portfolios.
- 2. Fuzzy Modeling:** Effective for smaller portfolios with significant uncertainty.
- 3. Decision-Aid Techniques (AHP/ANP):** Simple project ranking; limited by scalability.
- 4. Hybrid Models (Simheuristics):** Integrate simulation, fuzzy logic, and metaheuristics for dynamic and real-time decision-making.

“A CLUSTERING-BASED REVIEW ON PROJECT PORTFOLIO OPTIMIZATION METHODS”

Results

Study Results

Metaheuristics:

- Most effective for **large-scale portfolios** with realistic constraints.

Fuzzy Modeling:

- Better suited for **small, uncertainty-heavy portfolios**.

Hybrid Models (Simheuristics):

- Ideal for **dynamic project management** due to their:
 - **Adaptability and scalability.**
 - Ability to manage **multi-objective optimization** and **real-time decisions.**

Decision-Aid Techniques:

- Work well for project **ranking** in simpler scenarios but face **scalability limitations.**

Efficient Frontier Approaches:

- Enable decision-makers to explore **trade-offs** using Pareto optimality.

Identified Gaps	Description	Opportunities for Research
Real-Time Data Integration	Limited use of dynamic, real-time data inputs.	Develop models that integrate real-time decision-making and continuous data updates.
Scalability of Hybrid Models	Computational challenges for large portfolios.	Improve computational efficiency of simheuristics and hybrid methods.
Decision-Making in Dynamic Environments	Limited adaptability for evolving project scenarios.	Explore advanced adaptive algorithms for real-world, dynamic conditions.
Practical Applications	Models not fully validated across industries.	Apply PPO methods to diverse industries (e.g., healthcare, manufacturing, energy).

“INTEGRATED FRAMEWORK FOR PROJECT PORTFOLIO PLANNING MANAGEMENT UNDER RESOURCE CONSTRAINTS”

Synopsis

- This study addresses challenges in **project portfolio management (PPM)**, focusing on maximizing returns and optimizing resource allocation.
- Introduces an **integrated framework** that combines **planning** and **implementation** phases to improve outcomes.

Key Insight

Separating planning and implementation leads to suboptimal results due to uncertainties during execution.

Study Findings

Portfolio Methodologies:

1. Analyzed 23 papers grouped by **direction** (top-down vs. bottom-up) and **breadth** (planning vs. implementation).
2. **Top-down Approaches:** Emphasize interdependencies among projects; dominant in PPM studies.
3. **Bottom-up Approaches:** Treat projects independently; less explored.

Integrated Strategies:

1. Yielded better portfolio performance by addressing implementation uncertainties during planning.
2. Allocating **idle resources** in the planning phase mitigated execution disruptions and increased **earned portfolio value**.

“INTEGRATED FRAMEWORK FOR PROJECT PORTFOLIO PLANNING MANAGEMENT UNDER RESOURCE CONSTRAINTS”

Gaps in Study and Opportunities for Future Research

Identified Gaps	Description	Opportunities for Research
Integration of Planning and Implementation	Studies often focus on short-term gains without addressing execution uncertainties.	Develop models that integrate planning and execution phases under dynamic conditions.
Limited Bottom-Up Methodologies	Most research emphasizes top-down approaches.	Explore bottom-up methodologies or hybrid approaches combining top-down and bottom-up methods.
Practical Application and Validation	Framework lacks real-world testing and real-time adjustments.	Apply the framework in real-world environments , incorporating dynamic resource adjustments and evolving project scopes .

“ROBUST OPTIMIZATION FOR HURRICANE PREPAREDNESS”

Synopsis

- This study introduces a **robust optimization framework** for **hurricane preparedness**, focusing on minimizing **social costs** (logistics, deprivation, and fatalities).

Optimization Models

- **Single-Stage Adaptive Robust Model:** Determines optimal timing for deploying resources to Points of Distribution (PODs).
- **Two-Stage Robust Model:** Integrates **recourse actions** to adapt decisions on POD location, stockpile capacity, and resource flow.

Key Feature

Incorporates **deprivation costs** to quantify human suffering due to delayed relief, aligning with **humanitarian logistics** principles.

Model Components

Objective Function: Minimize total travel cost for disaster relief

$$\text{Minimize } \sum_{i \in I} \sum_{j \in J} \sum_{s \in S} d_{ij} h_{is} Y_{ijs}$$

Facility Capacity Constraint

$$z_{jt} \geq b_{jt} \cdot X_{jt}, \quad z_{jt} > B_{jt} \cdot X_{jt}, \forall i \in I, \forall j \in J, \forall s \in S$$

Closest Assignment Constraint

$$\sum_{k \in J | d_{ik} \leq d_{ij}} Y_{ijs} \leq X_{jt}, \quad \forall i \in I, \forall j \in J, \forall s \in S, \quad \forall t \in S | t \geq s$$

Maximum Travel Distance Constraint

$$Y_{ijs} = 0, \forall i \in I, \forall j \in J, \forall s \in S | d_{ij} > D_{is}$$

Single Decision Variable Assignment Constraint

$$Y_{ijs} \in \{0,1\}, X_{ijs} \in \{0,1\}, \quad z_{jt} \geq 0, \forall i \in I, \forall j \in J, \forall s \in S, t \in S$$

“ROBUST OPTIMIZATION FOR HURRICANE PREPAREDNESS”

Study Results

Comparison of Models (CM vs. HCM):

- 1. HCM Model:
 - 1. Placed **16 facilities** in both scenarios (40% and 50% population staying behind).
 - 2. Improved **supply-to-person ratio** from 1.6 (CM) to **1.3** (HCM).
 - 3. Lowered average travel costs for **Type III services** (lower-level demands).

Challenges:

- 1. HCM computational time: Up to **93.59 hours**, limiting large-scale practicality.

Computational Performance:

- 1. CM Model: Faster but less resource-efficient.
- 2. HCM Model: More efficient resource allocation but requires computational advancements for scalability.

Gaps in Study and Opportunities for Future Research

Identified Gaps	Description	Opportunities for Research
Computational Complexity	HCM requires extensive computational time.	Develop faster algorithms to improve scalability for large-scale implementations.
Dynamic Resource Allocation	Static assumptions for POD locations and capacities.	Explore adaptive frameworks to respond to real-time changes during disasters.
Incorporating Real-Time Data	Limited integration of dynamic, real-time data.	Integrate real-time demand forecasts for more responsive disaster planning.
Broader Applications	Focused solely on hurricane preparedness.	Apply the robust optimization framework to other disaster scenarios (e.g., earthquakes, floods).

“OPTIMIZATION OF RENEWABLE ENERGY PROJECT PORTFOLIO SELECTION USING HYBRID AIS-AFS ALGORITHM IN AN INTERNATIONAL CASE STUDY”

Synopsis	Key Contributions
Focus: Optimizing Renewable Energy Project Portfolios (REPPs) to maximize net present value (NPV) and minimize investment risk. Methodology: - Introduced a hybrid algorithm (HFSIS) combining Artificial Immune System (AIS) and Artificial Fish Swarm (AFS) methods. - Evaluated 137 projects from 20 countries with various portfolio sizes (3, 5, 10, 15 projects). Minimize Risk: Evaluates financial variability across projects. Efficient Frontier Visualization (EIF): - Facilitates decision-making by showcasing risk-return trade-offs.	Dataset Reliability: - Sourced from reputable repositories (e.g., NREL, World Bank) for global representation. Mathematical Model: Multi-objective optimization with two key functions: Maximize NPV: Captures profitability over project lifetimes. Minimize Risk: Evaluates financial variability across projects. Efficient Frontier Visualization (EIF): - Facilitates decision-making by showcasing risk-return trade-offs.

“OPTIMIZATION OF RENEWABLE ENERGY PROJECT PORTFOLIO SELECTION USING HYBRID AIS-AFS ALGORITHM IN AN INTERNATIONAL CASE STUDY”

Study Results and Analysis

Strengths

1. Algorithm Performance:

1. HFSIS outperformed standalone AIS and AFS methods in profitability and risk management.
2. Adaptable to various portfolio sizes with consistent global diversification.

2. Sensitivity Analysis:

1. Demonstrated impact of **initial investment (IO)** and **interest rates (r)** on NPV:
 1. **IO Decrease (30%)**: Made NPV positive across all projects.
 2. **Interest Rate Changes**: Highlighted non-linear impacts on profitability.

3. Realistic Constraints:

1. Incorporates practical factors like loan repayments, workforce costs, and project lifetimes.

Weaknesses

1. Dataset Limitations:

1. Small-scale validation; lacks large-scale, real-world testing.

2. Static Assumptions:

1. Fixed interest/inflation rates limit adaptability to dynamic markets.

3. Parameter Sensitivity:

1. Hybrid algorithm highly dependent on parameter tuning (e.g., cloning rate, visual radius).

“OPTIMIZATION OF
RENEWABLE ENERGY PROJECT
PORTFOLIO SELECTION USING
HYBRID AIS-AFS ALGORITHM
IN AN INTERNATIONAL CASE
STUDY”

Gaps in Study and Opportunities
for Future Research

Identified Gaps	Description	Proposed Actions
Model Validation	Validate the model on larger, real-world datasets and dynamic markets.	Test with diverse datasets , including varying portfolio sizes and regions.
Dynamic Variables	Static interest/inflation rates reduce adaptability.	Incorporate time-varying parameters to reflect real-world market fluctuations.
Sensitivity Analysis Expansion	Limited to IO and interest rates; lacks multi-factor interactions.	Conduct multi-factor analyses to understand parameter interactions on portfolio outcomes.
Algorithm Efficiency	Parameter tuning impacts hybrid algorithm performance.	Use machine learning for tuning parameters like step size and visual radius.
Geographic Scope	Underrepresentation of small-scale or emerging market projects.	Expand datasets to include local and small-scale renewable energy projects .
Regulatory and Environmental Variables	Limited consideration of policies or environmental risks.	Include policy impacts (e.g., subsidies, regulations) and environmental constraints.
Algorithm Comparison	Lack of benchmarking against advanced methods.	Compare HFSIS with state-of-the-art optimization techniques to validate efficiency.
Real-World Case Studies	Limited application to practical scenarios.	Validate models with real-world projects to assess feasibility and performance.

SIMILARITIES ACROSS STUDIES

Reliance on Optimization Techniques

Common Themes:

- Use of **multistage stochastic programming** and **risk-averse approaches** across studies.
- Effective in addressing **uncertainties** in project scheduling, resource allocation, and supply chain logistics.

Applications:

- Widely used in **construction, healthcare, and modular supply chains**, where **scenario-based decision-making** improves resource distribution.

Examples:

- **Risk-Averse Supply Chain for Modular Construction** applied risk-averse methods to unpredictable supply chains.
- **Risk-Averse Optimization and Control** proposed adaptive strategies for real-time adjustments.

SIMILARITIES ACROSS STUDIES (CONTINUED)

Multi-Skilled Human Resource Management

Key Insights:

- Managing **multi-skilled human resources** across projects improves efficiency.
- Integration of **internal and external resources** supports sectors with fluctuating demands (e.g., construction, healthcare).

Techniques Used:

- **Scenario Reduction:** Essential for managing large-scale optimization problems.
- **Applications:** Time-sensitive models in healthcare, construction, and disaster response.

SIMILARITIES ACROSS STUDIES (CONTINUED)

Shared Techniques and Goals

Scenario-Based Decision-Making

- **Key Points:**
 - Widely adopted across studies to anticipate future outcomes.
 - Ensures **adaptive responses** to changing conditions.

Examples:

- **Hurricane Disaster Relief Goods Distribution:** Evaluated resource needs under potential disaster scenarios.
- Improves efficiency in disaster response, construction, and logistics.

Portfolio Management

- **Optimization Focus:**
 - Leveraging **hybrid algorithms** (e.g., AIS-AFS) for renewable energy portfolios.
 - **Shared Themes:** Multi-objective optimization for profitability and risk minimization in uncertain environments.

Examples:

- Hybrid models like the AIS-AFS (Section 3.I) align with multistage stochastic programming models (Section 3.A).
- Both incorporate **risk minimization** as a core objective, showcasing adaptability across fields.

DIFFERENCES ACROSS STUDIES: DIFFERENCES IN METHODOLOGIES ACROSS STUDIES

Adaptability to Real-Time Changes

Disaster Response Models:

- **Highly adaptive** to real-time changes (e.g., weather, infrastructure damage).
- Example: **Hurricane Preparedness Models** dynamically update inputs for immediate decisions.

Traditional Project Management Models:

- Operate on **periodic updates**, assuming fixed resources.
- Example: **Multistage Stochastic Programming** models lack flexibility for sudden changes like budget cuts or delays.

Modular Construction Projects:

- Moderate adaptability through **planned updates** but lack real-time responsiveness

DIFFERENCES ACROSS STUDIES: DIFFERENCES IN RISK MANAGEMENT APPROACHES

Risk Management

Risk-Neutral Models:

- Focus on **financial risks** like cost overruns; assume stable conditions.
- Example: Models in **traditional project management**.

Risk-Averse Models:

- Include **dynamic risks** (e.g., workforce changes, stakeholder demands).
- Example: **Risk-Averse Optimization and Control** emphasizes proactive decision-making.

Hybrid Models in Healthcare and Construction:

- Mix **risk-neutral** and **risk-averse** strategies to manage supply chain uncertainties.

DIFFERENCES ACROSS STUDIES: RESOURCE STRATEGIES AND HANDLING NON-FINANCIAL RISKS

Resource Deployment Strategies

- **Disaster Scenarios:**
 - Start with **broad resource deployment** to prioritize availability and minimize response delays.
- **Traditional Project Management:**
 - Start with **minimal resources**, scaling up based on needs to maximize cost efficiency.

Non-Financial Risks

- **Traditional Models:**
 - Focus on **financial risks**, often neglecting factors like employee turnover or regulatory shifts.
- **Sector-Specific Models (Healthcare, Modular Construction):**
 - Address **multi-skilled human resource management** but struggle with broader uncertainties (e.g., market changes).

DIFFERENCES ACROSS STUDIES: COMPUTATIONAL EFFICIENCY AND SCENARIO REDUCTION

Computational Efficiency

Large-Scale Optimization (Modular Construction):

- Prioritize **scenario reduction** for managing large datasets efficiently.
- Example: **Stochastic Optimization Models**.

Dynamic Resource Allocation Models:

- Optimize resource use over time but focus less on immediate computational speed.

Renewable Energy Optimization (AIS-AFS):

- Balances **financial and environmental considerations**, integrating global risks and opportunities.

FUTURE RESEARCH OPPORTUNITIES AND POSSIBLE AREAS OF FOCUS

Advancing Risk-Averse Project Management

- **Core Objective:** Enhance project selection and scheduling methods under uncertain conditions using risk-averse modeling.

Key Areas of Focus:

- **Dynamic, Multistage Stochastic Programming:**
 - Adapt models to changing conditions using iterative, stage-wise decision-making.
- **Time-Consistent Risk Measures:**
 - Continuously update risk evaluations based on new information.
 - Address low-probability, high-impact events for greater resilience.
- **Integration of Time Factors:**
 - Explore **Markov risk-averse frameworks** for balancing time-to-completion with profit maximization.
- **Scalability:**
 - Expand models to handle larger project portfolios while maintaining risk aversion.

FUTURE RESEARCH IMPACT

Goals and Contributions

- **Adaptive Strategies:** Develop models that support real-time adjustments to risk and resource changes.
- **Resilience:** Ensure projects are robust against disruptions and uncertainties.
- **Applicability:** Apply these models across sectors, enhancing decision-making and resource optimization.

CONCLUSION

Key Takeaways from Literature Review

- **Advancements in Methodologies:**
 - Techniques like **multistage stochastic programming** and **risk-averse approaches** enable decision-makers to navigate uncertainty effectively.
- **Remaining Gaps:**
 - Need for **comprehensive risk models**, **time-dependent resource allocation**, and **dynamic decision-making integration**.

Focus of Future Research

- Build on the foundation of **A Multistage Stochastic Programming Approach in Project Selection and Scheduling**.
- Develop:
 - **Dynamic, Multistage Models:** Adapt to changing conditions.
 - **Time-Consistent Risk Measures:** Enable iterative, real-time decision-making.
 - **Integration of Markov Frameworks:** Balance time-to-completion and profit.
- Expand research to manage larger project portfolios, ensuring both risk aversion and scalability.

Impact:

- Contribute to **adaptive, resilient project management strategies**.
- Enhance decision-making under uncertainty, leading to **better project outcomes across sectors**.

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Q & A

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